

Assignment 4 Report

English $\rightarrow$ Hindi

Roll No: **M25CSA025**

1 Baseline Results

Description: The baseline model was trained for 100 epochs without hyperparameter tuning.

Metrics:

- Training Time: **140 min**
- Final Loss: **0.0982**
- BLEU Score: **55.25**

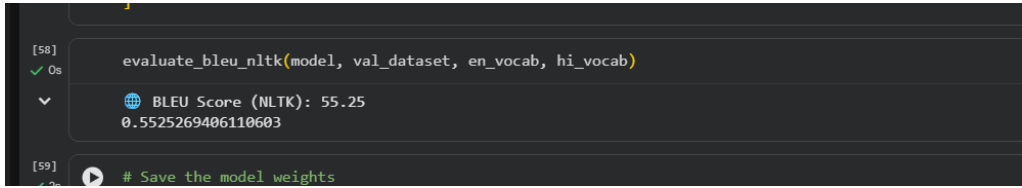


Figure 1: Baseline Training Output

2 Hyperparameter Tuning

Tuned Hyperparameters and Ranges:

Hyperparameter	Search Range
Learning Rate	(1e-5,1e-3)
Batch Size	[16,32,34]
Number of Heads	[4,8]
Feedforward Dimension (d_{ff})	[1024,2048]
Dropout	(0.1,0.4)

```

train_tune_b500589b RUNNING 8.57614e-05 32 8 1024 0.159189 2 120.727 4.10706 |
train_tune_6808935b TERMINATED 1.46094e-05 16 8 2048 0.220668 3 259.055 4.63165 |
train_tune_fc2940a2 TERMINATED 0.000133677 32 4 1024 0.391689 3 176.896 4.10444 |
train_tune_d11f4308 TERMINATED 0.000220027 16 8 2048 0.297771 3 269.535 4.51297 |
train_tune_942c145e TERMINATED 0.000499534 32 8 2048 0.32716 3 227.141 5.63775 |
2026-03-17 13:50:36,024 INFO tune.py:1009 -- Wrote the latest version of all result files and experiment state to '/root/.ray_results/train_tune_2026-03-17_13-31-21' in 0.0071s.

Trial train_tune_b500589b completed after 3 iterations at 2026-03-17 13:50:36. Total running time: 19min 12s
-----+-----+
| Trial train_tune_b500589b result |
+-----+-----+
| checkpoint_dir_name |
| time_this_iter_s | 59.0439 |
| time_total_s | 179.771 |
| training_iteration | 3 |
| loss | 3.5772 |
+-----+-----+

Trial status: 5 TERMINATED
Current time: 2026-03-17 13:50:36. Total running time: 19min 13s
Logical resource usage: 2.0/2 CPUs, 1.0/1 GPUs (0.0/1.0 accelerator_type:t4)
-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Trial name | status | lr | batch_size | num_heads | d_ff | dropout | iter | total time (s) | loss |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| train_tune_6808935b | TERMINATED | 1.46094e-05 | 16 | 8 | 2048 | 0.220668 | 3 | 259.055 | 4.63165 |
| train_tune_fc2940a2 | TERMINATED | 0.000133677 | 32 | 4 | 1024 | 0.391689 | 3 | 176.896 | 4.10444 |
| train_tune_d11f4308 | TERMINATED | 0.000220027 | 16 | 8 | 2048 | 0.297771 | 3 | 269.535 | 4.51297 |
| train_tune_942c145e | TERMINATED | 0.000499534 | 32 | 8 | 2048 | 0.32716 | 3 | 227.141 | 5.63775 |
| train_tune_b500589b | TERMINATED | 8.57614e-05 | 32 | 8 | 1024 | 0.159189 | 3 | 120.727 | 4.10706 |

```

Figure 2: Hyperparameter Tuning Process

3 Best Configuration

The best hyperparameter configuration obtained from tuning is:

- Learning Rate: **8.576135093769627e-05**
- Batch Size: **32**
- Number of Heads: **8**
- Feedforward Dimension (d_{ff}): **1024**
- Dropout: **0.15918892050996541**

```

best_result = results.get_best_result(metric="loss", mode="min")

best_config = best_result.config

print("Best configuration:", best_config)

Best configuration: {'lr': 8.576135093769627e-05, 'batch_size': 32, 'num_heads': 8, 'd_ff': 1024, 'dropout': 0.15918892050996541}

```

Figure 3: Best Hyperparameter Configuration

4 Final Model Results

The model was retrained using the best hyperparameter configuration obtained from tuning. Two experiments were conducted to analyze performance improvement with increased training:

Run 1: 20 Epochs

Metrics:

- Training Time: **20 min**
- Final Loss: **0.4613**
- BLEU Score: **53.87**

Observation: The BLEU score obtained after 20 epochs was approximately similar to the baseline model, indicating that the model had not yet fully benefited from the tuned hyperparameters.

```
[72] ✓ 20m train(model, train_loader, optimizer, criterion)

... Checkpoint saved at epoch 1, loss 5.1568.
Checkpoint saved at epoch 2, loss 4.0961.
Checkpoint saved at epoch 3, loss 3.5684.
Checkpoint saved at epoch 4, loss 3.1617.
Checkpoint saved at epoch 5, loss 2.8180.
Checkpoint saved at epoch 6, loss 2.5189.
Checkpoint saved at epoch 7, loss 2.2543.
Checkpoint saved at epoch 8, loss 2.0070.
Checkpoint saved at epoch 9, loss 1.7847.
Checkpoint saved at epoch 10, loss 1.5902.
Checkpoint saved at epoch 11, loss 1.4126.
Checkpoint saved at epoch 12, loss 1.2387.
Checkpoint saved at epoch 13, loss 1.0919.
Checkpoint saved at epoch 14, loss 0.9662.
Checkpoint saved at epoch 15, loss 0.8459.
Checkpoint saved at epoch 16, loss 0.7446.
Checkpoint saved at epoch 17, loss 0.6628.
Checkpoint saved at epoch 18, loss 0.5787.
Checkpoint saved at epoch 19, loss 0.5125.
Checkpoint saved at epoch 20, loss 0.4613.
```

Figure 4: Final Model Training (20 Epochs)

```
[80] ✓ 0s evaluate_bleu_nltk(model, val_dataset, en_vocab, hi_vocab)

🌐 BLEU Score (NLTK): 53.87
0.5387044251994753

[] Start coding or generate with AI.
```

Figure 5: Bleu score (20 Epochs)

Run 2: 30 Epochs

Metrics:

- Training Time: **30 min**
- Final Loss: **0.2389**
- BLEU Score: **67.79**

Observation: After increasing the number of epochs to 30, the model showed noticeable improvement. The BLEU score increased compared to both the baseline and the 20-epoch run, demonstrating that the tuned configuration required slightly longer training to realize its full potential.

```
[91]
✓ 30m
train(model, train_loader, optimizer, criterion)

Checkpoint saved at epoch 1, loss 5.1714.
Checkpoint saved at epoch 2, loss 4.1098.
Checkpoint saved at epoch 3, loss 3.5708.
Checkpoint saved at epoch 4, loss 3.1616.
Checkpoint saved at epoch 5, loss 2.8210.
Checkpoint saved at epoch 6, loss 2.5299.
Checkpoint saved at epoch 7, loss 2.2543.
Checkpoint saved at epoch 8, loss 2.0151.
Checkpoint saved at epoch 9, loss 1.7885.
Checkpoint saved at epoch 10, loss 1.5986.
Checkpoint saved at epoch 11, loss 1.4170.
Checkpoint saved at epoch 12, loss 1.2480.
Checkpoint saved at epoch 13, loss 1.0988.
Checkpoint saved at epoch 14, loss 0.9683.
Checkpoint saved at epoch 15, loss 0.8530.
Checkpoint saved at epoch 16, loss 0.7494.
Checkpoint saved at epoch 17, loss 0.6573.
Checkpoint saved at epoch 18, loss 0.5852.
Checkpoint saved at epoch 19, loss 0.5214.
Checkpoint saved at epoch 20, loss 0.4573.
Checkpoint saved at epoch 21, loss 0.4162.
Checkpoint saved at epoch 22, loss 0.3871.
Checkpoint saved at epoch 23, loss 0.3474.
Checkpoint saved at epoch 24, loss 0.3154.
Checkpoint saved at epoch 25, loss 0.2999.
Checkpoint saved at epoch 26, loss 0.2868.
Checkpoint saved at epoch 27, loss 0.2675.
Checkpoint saved at epoch 28, loss 0.2567.
Checkpoint saved at epoch 29, loss 0.2451.
Checkpoint saved at epoch 30, loss 0.2389.
```

Figure 6: Final Model Training (30 Epochs)

```
[99]
✓ 0s
evaluate_bleu_nltk(model, val_dataset, en_vocab, hi_vocab)

BLEU Score (NLTK): 67.79
0.6778564842860517
```

Figure 7: Bleu score (30 Epochs)

5 Comparison with Baseline

Metric	Baseline	Tuned Model
Total epochs	100	30
Training Time(min)	140	30
Total loss	0.0982	0.2389
BLEU Score	55.25	67.79

6 Best model

Saved model is uploaded on huggingface the link is attached here.

Link: <https://huggingface.co/kartikIyer27/model-en-to-hi-mlops/tree/main>

7 GitHub Repository

The complete implementation, including source code, trained model, vocabulary files, and experiment results, has been uploaded to GitHub.

Link: <https://github.com/KartikIyer27/MlOps-kartik-m25CSA025/tree/Assignment-4>